



Administrativa

13 October 2025

Prof. Dr. Sebastian Wild

Goals for Today

- ▶ give you some detail on **what** this module covers

⇒ so that you can decide whether to keep it

(if it is an elective module for you)

Efficient Algorithms / Effiziente Algorithmen

- ▶ inform you about **how** EA is run
- ▶ inform you about how EA is **assessed**

Welcome to CS 566 – Efficient Algorithms

► Dozent: Prof. Dr. Sebastian Wild
Mehrzweckgebäude, Raum 05 D 16
`wild@informatik.uni-marburg.de`

Betreuer: Tamio-Vesa Nakajima

Tutors: Finn Moltmann `moltmann@students.uni-marburg.de`
Arsenij Winkel `winkelar@students.uni-marburg.de`

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- Module website: www.wild-inter.net/teaching/ea

→ your first address for any infos on CS 566

- *Campuswire*: collaborative Q&A (more on this later)

also used for announcements

→ please register asap (link also on ILIAS)

<https://campuswire.com/p/G66349A19>

PIN 1794

- *Slido*: student response system for formative feedback → bring a smart device to class!

- Final mark: 100% final exam (Klausur)

Zulassungsvoraussetzungen zur Klausur: 50% of points from exercise sheets



A Note on Languages

- ▶ Module is mostly in German
 - ▶ in particular examinations
 - ▶ except as prerequisite for English M.Sc. admission
If that's you, stay tuned.
I'll come to that!
- ▶ some written material in English
 - ▶ in particular slides
- ▶ Why?
 - ▶ English is the *lingua franca* of our time
~> you profit from exposure
 - ▶ people (=future employers!) will assume you can at least read English
 - ▶ in young computer science, technical terms are already English
- ▶ Also, it's 2025! AI tools bridge lots of language gaps 🐱
DeepL, Google Translate, ChatGPT etc.



CS 566 for Credit vs. for Conditional Admission

▶ (Normal / for-credit version of) CS 566:

- ▶ Taken by students in various undergrad or masters programs
- ▶ Compulsory for German *B.Sc. Data Science*
- ↪ Offered in German (including exams)

▶ CS 566 for conditional admission (into *M.Sc. Data Science*):

- ▶ full program in English, international students
- ↪ Separate English examinations

- ▶ formally separate from CS 566
- ▶ examination is pass/fail only
- ▶ **If required for admission, you cannot also take CS 566 for credit.**

- ▶ Examination based on English self-study materials (not full lectures) ↪ module website
- ▶ Welcome to attend lectures, and tutorials (space permitting) 
- ▶ Join the Campuswire Q&A and team up with others to study!

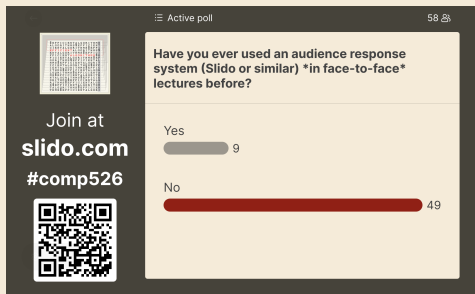
Audience Response System: *Slido*

- ▶ Goal: Collect immediate, formative feedback
 - ▶ Stay focused and engaged! (“active learning”)
 - ▶ Quick feedback (for you individually) if you are on track.
 - ▶ Quick feedback (for me) whether (most of) you are on track.

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- ▶ Slido has 2 useful features:

1. Quick Polls



The screenshot shows a Slido poll interface. On the left, there is a QR code and the text "Join at slido.com #comp526". The main area displays the poll question: "Have you ever used an audience response system (Slido or similar) *in face-to-face* lectures before?". Below the question, there are two horizontal progress bars. The "Yes" bar is grey and shows 9 responses. The "No" bar is red and shows 49 responses. The top of the interface indicates "Active poll" and "58" participants.

Join at
slido.com
#comp526

Active poll 58

Have you ever used an audience response system (Slido or similar) *in face-to-face* lectures before?

Yes 9

No 49

2. Audience Questions



The screenshot shows a Slido Q&A interface. On the left, there is a QR code and the text "Join at slido.com #comp526". The main area displays a list of questions. The first question is from "Sebastian Wild" and asks "How can I ask a question in class?". The second question is from "Anonymous" and asks "I'm a bit unsure, I'd rather ask this anonymously.". The top of the interface indicates "Q&A", "Popular", and "2" questions.

Join at
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Q&A Popular 2

Sebastian Wild
How can I ask a question in class?

Anonymous
I'm a bit unsure, I'd rather ask this anonymously.

Clicker Question



Have you used an audience response system (Slido or similar) in lectures before?

A Yes

B No



→ *slido.do/cs566*

My approach to lectures

My conclusions (from years of own experience, a pandemic, and observing others)

irrespective of the
mode of delivery!



0. Good explanations (intuitions!) and well-structure material are the most important aspect.
1. **Synchronous (live) lectures** beat videos in keeping up with class. (but recordings are great!)
2. Only a small minority of students asks questions in class. ~~~ other backchannels
3. **Interaction** makes content memorable (and keeps brains awake!) ~~~ *Slido* tasks

Components of EA

Slido questions

immediate feedback
simple questions

Lectures

new material
discussions
big picture

Tutorials

get practice solving problems
solve deep questions

Campuswire

collaborative Q&A knowledge base

Exam Question Gallery

collaborative pool of potential and past exam problems

Final Exam

summative assessment
of your acquired skills

Overview of the module

Goals:

- ▶ build / enhance your toolbox of algorithmic methods and techniques
 - ↪ here: focus on practical methods
- ▶ enable you to reason about and communicate algorithmic solutions
 - ↪ level of abstraction, proofs, mathematical analysis, vocabulary
- ▶ enable you to apply, combine and extend methods

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Units: (0. Administrative)

- | | |
|--------------------------------|---------------------------|
| 1. Proof Techniques | 8. Clever Codes |
| 2. Machines & Models | 9. Graph Algorithms |
| 3. Fundamental Data Structures | 10. Parallel Algorithms |
| 4. Efficient Sorting | 11. Greedy Algorithms |
| 5. Divide & Conquer | 12. Dynamic Programming |
| 6. String Matching | 13. Text Indexing |
| 7. Text Compression | 14. Range-Minimum Queries |

Assessments

- ▶ **Module mark** = mark in final written exam
- ▶ **Final exam**
 - ▶ written examination
 - ▶ Dates tbd Likely: 18 Feb and 25 March
- ▶ To pass the module, you have to pass either of the exams
 - ▶ If you pass the first exam, you *cannot* take the second to improve you mark
- ▶ **Exam Material:** everything covered in lectures (except marked “~~✗~~ exam”)
everything covered in **tutorials** and exercise sheets
- ▶ **Admission requirements to final exam**
 - ▶ ≤ 2 exercise sheets with 0 points in your group
(not handing in implies 0 points)
 - ▶ $\geq 50\%$ of available points in sum over all exercise sheets
 - ▶ We plan with 12 marked exercise sheets in total

stay tuned ...

Tutorials

- ▶ *Exercise Sheet* (Übungsblatt)
 - ▶ released on module website every week
 - ▶ to be **handed in**
 - ▶ online on ILIAS
 - ▶ in **groups** of 3 students
 - ▶ roughly one week to work on it
 - ▶ practice problems (some old exam questions, too!)
 - ▶ enhancement problems
- ▶ in *tutorials*
 - ▶ discussion of solutions (in the week after hand-in)
 - ▶ work on **in-class exercises** (Präsenzaufgaben)
 - ▶ to prepare you for next marked exercise sheet
 - ▶ *not* handed in or marked

*Use the tutorials to **practice your thinking!*** = Don't cheat yourself!

*"If I tell you to run 10km,
it isn't because I want you
to be 10km away from me."*

Generative AI

We live in exciting times!

LLMs (ChatGPT etc.), Media generators
(Midjourney etc.), GitHub CoPilot, ...

- ▶ Generative Artificial Intelligence (GenAI) is amazing!
 - ▶ full of flaws (hallucination, bias, copyright, data privacy, cost, ...)
 - ▶ and yet ... often helpful, surprisingly versatile
- ▶ Why not use for everything?
 - ▶ Need for *deeply skilled* humans here to stay (for now anyways)
 - ~> **Skill comes from practice!** (We still teach mental arithmetic in primary school!)



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assessments designed for upskilling *humans*

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Acceptable use:

- ▶ preparatory research
(≈ Wikipedia)
- ▶ proof reading
(spelling, grammar)

Unacceptable use: (not exhaustive!)

- ▶ use generated parts w/o acknowledgment & citation
- ▶ tools to paraphrase others' work to pass as own
- ▶ generated parts with inappropriate prompt,
e. g., "write me a conclusion for this essay"

Clicker Question



What do you think is the #1 **predictor** of whether a student cheats in assessments?



→ *sli.do/cs566*

Clicker Question



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Source: youtu.be/sMpC8QwWSbI

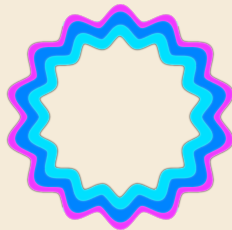


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What is Campuswire?

Campuswire is an online space for lectures

1. **Class Feed**: questions on material
2. **Chatrooms**: structured social space
similar to Slack or Discord



Join via link on website:
campuswire.com/p/G66349A19

Use in browser
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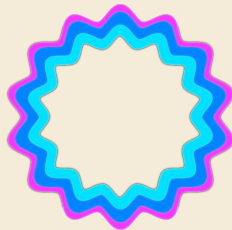
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We use Class Feed for **collaborative Q&A**

- ▶ Ask **public** questions
 - ▶ “Why is $\lg(n^3) = \Theta(\log n)$?”
 - ▶ “Will there be classes on regional holiday X?”
- ▶ **Answer your peers’ questions!**
 - ▶ Know the answer? → put it in!
 - ▶ Know a partial answer? → Post it, others can build on it!
 - ▶ Found a helpful answer (or question)? → Vote it up!
- ▶ Ask **private** questions
 - ▶ if your question might contain “spoilers” for assessments
 - ▶ if you feel the answer is only relevant for you personally

How to Campuswire

► Our goals for Campuswire Q&A:

1. **be fair** Same answers for everyone
2. **learning by teaching** YOU will answer most questions!
3. **be inclusive** posts can be anonymous; you can take your time to ask and answer

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- ▶ Therefore, we instructors will
 - ▶ redirect you to Class Feed for questions,
 - ▶ wait before answering, to give other students a chance to answer first,
 - ▶ explicitly mark good answers (and questions!) as such

ILIAS

- ▶ Official announcements
- ▶ Hand-in of exercise sheets
- ▶ Announcement of marks

... what can be on the public module website
goes to the public module website!



Exam Question Gallery

- ▶ We jointly collect a **pool of exemplary exam questions**.
- ▶ *You add **your** questions to it.*
- ▶ I will give feedback which questions are realistic.
- ▶ *... and we will pick one if there's sufficiently many good ones!*

↪ great resource for exam preparation

↪ We will answer selected questions in recap session (last week of classes)

- ▶ Engage in this early and pose great questions

- ▶ Start today: <https://tiny.cc/ea-exam-question-gallery>

Philosophy of the module

CS 566 is part of a *scientific* course.

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Less ...



<https://imgur.com/gallery/vX118>

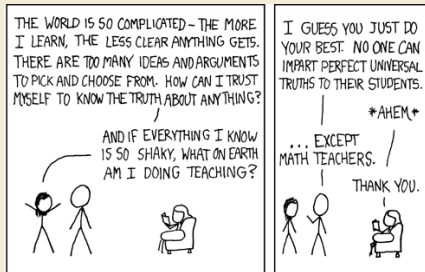
Philosophy of the module

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Less ...



... and more



Philosophy of the module

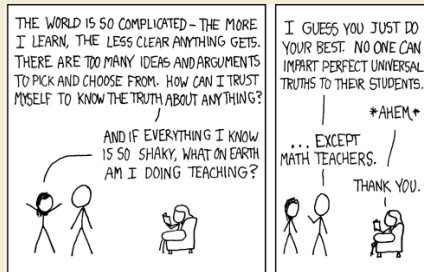
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<https://xkcd.com/263/>

~> Focus on *universal truths* of practical algorithms

- ▶ model of reality (machines, programs, data)
- ▶ quantitative predictions
- ▶ validate model in experiments

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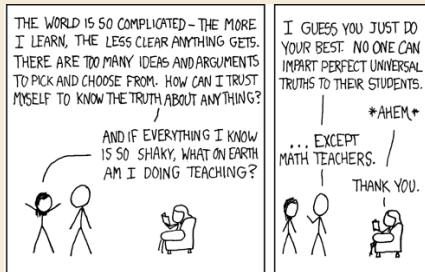
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~> Need some math techniques. (up next)